



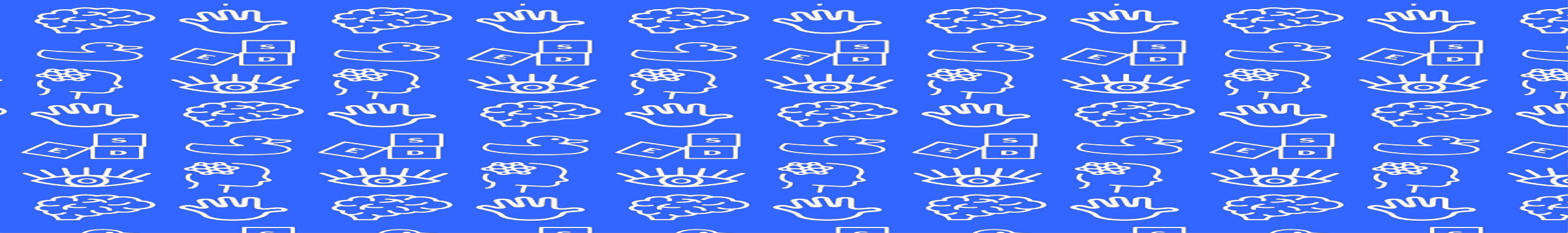
South Carolina

Caregiver Screening Attitudes and Data Quality

A plain language walk through of what we found

Early Social Development Lab, University of South Carolina

Based on 1,956 survey records. Same numbers as the full technical deck, fewer of them on each slide.



What we did, in one slide

We asked caregivers how they feel about infant autism screening. Then, separately, we checked whether those feelings line up with what they say they would actually do.

WHAT BUILDS THE GROUPS

10 attitude questions only

Trust in clinicians, comfort with a blood test, whether screening feels ethical, how much distress it would cause, how useful it seems, tolerance for a wrong answer, openness to equipment, feasibility, low burden options, and how unpleasant it feels.



KEPT OUT, CHECKED AFTERWARDS

4 things we never let in

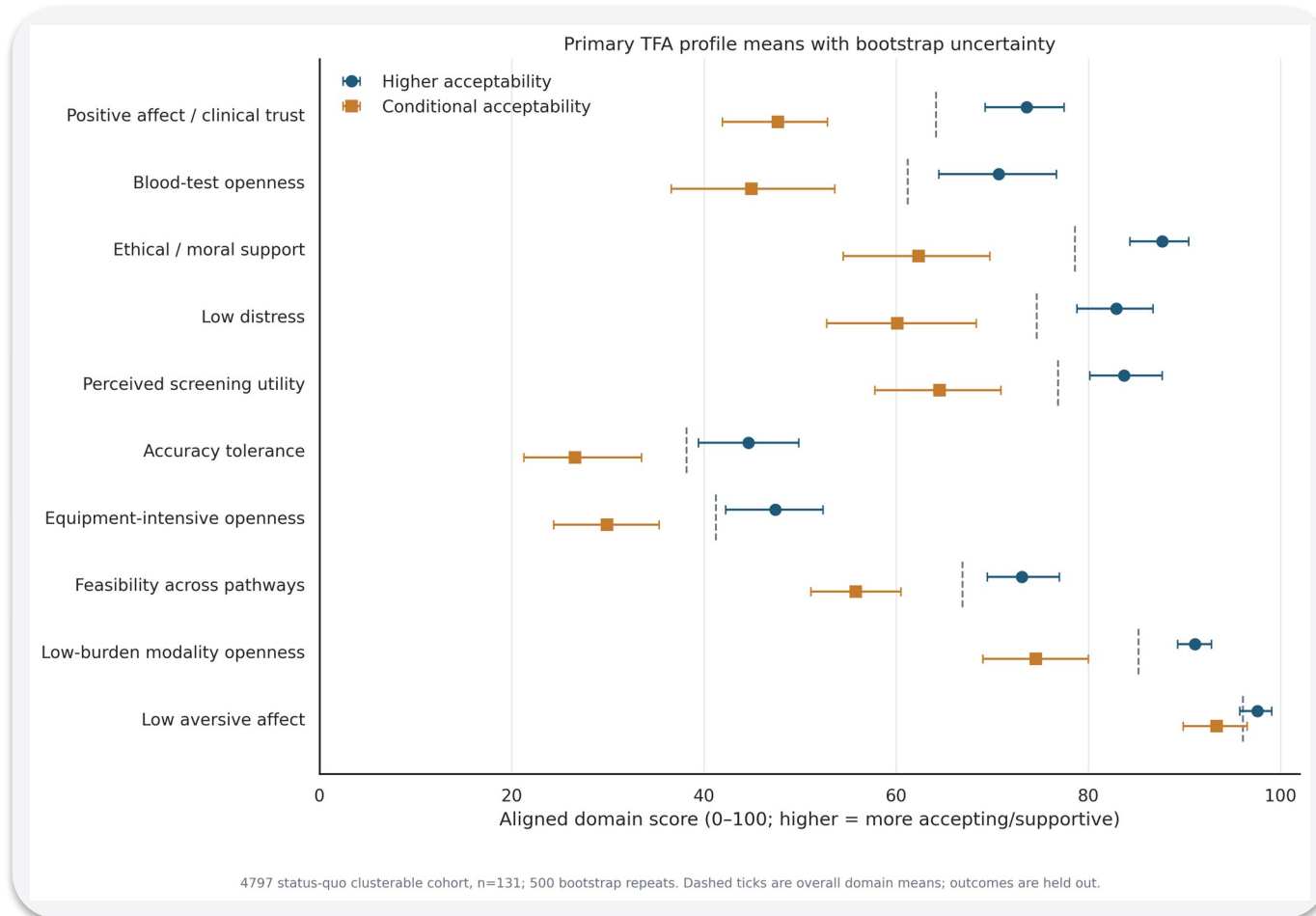
- **Screening intent** Would you get your baby screened?
- **Autism knowledge** A 7 question quiz with right answers
- **Personal values** What matters to you in life
- **Demographics** Education, work, area, family setup

HOW TO READ THIS

- 1 1,956 records came in. 131 caregivers answered enough attitude questions to be grouped. 127 of those also answered the screening question.
- 2 The screening answer never went into the grouping, so when we compare the groups on it later we are not arguing in a circle.
- 3 Everything on the right is a check we run after the groups already exist. It cannot bend the groups to fit a story.

Caregivers fall into two broad groups

One group is warm about screening across the board. The other is supportive too, but with conditions attached.

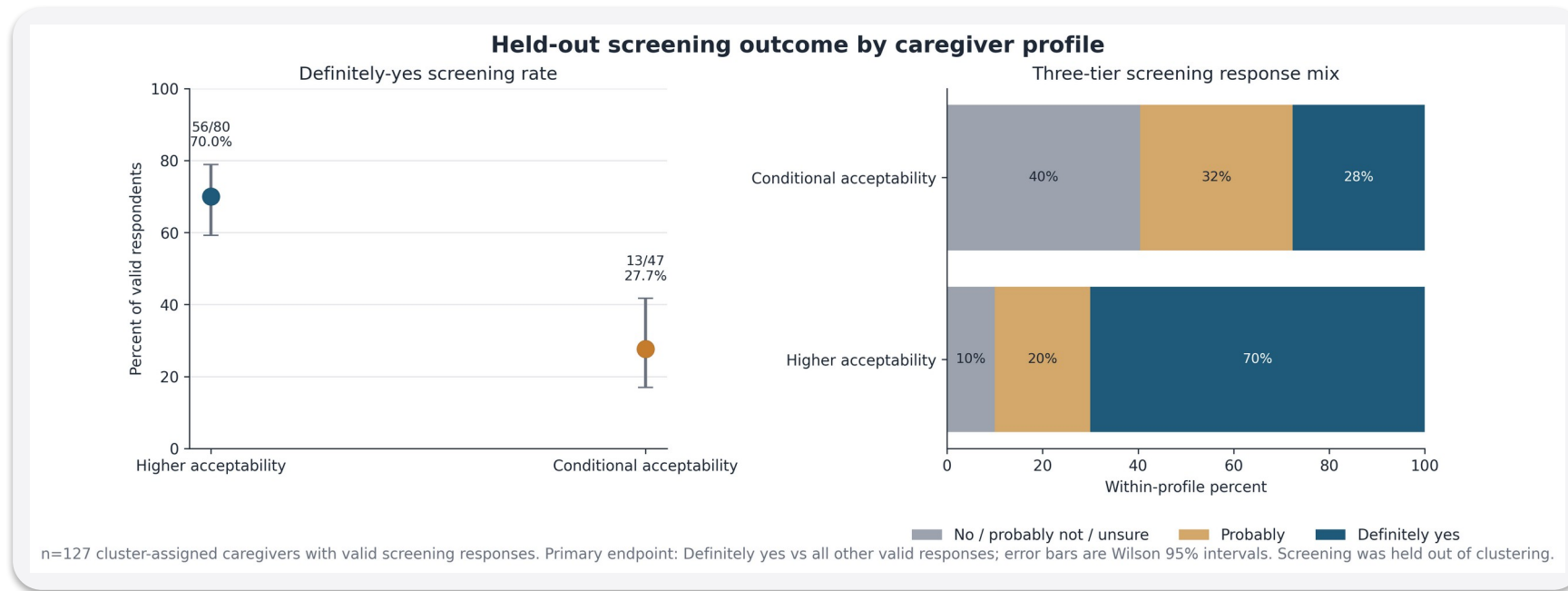


HOW TO READ THIS

- 1 Each row is one attitude. Further right means more supportive. Blue circles are the warmer group, orange squares are the conditional group.
- 2 The wider the gap between the two markers, the more that attitude separates the groups. The three widest gaps are all about trust and ethics.
- 3 Look at the bottom row. Both groups sit at the far right, so neither of them finds the screen itself upsetting. The disagreement is about the conditions around it.

The two groups answer the screening question very differently

About 7 in 10 caregivers in the warmer group said they would definitely screen their baby. In the conditional group it was closer to 3 in 10.



HOW TO READ THIS

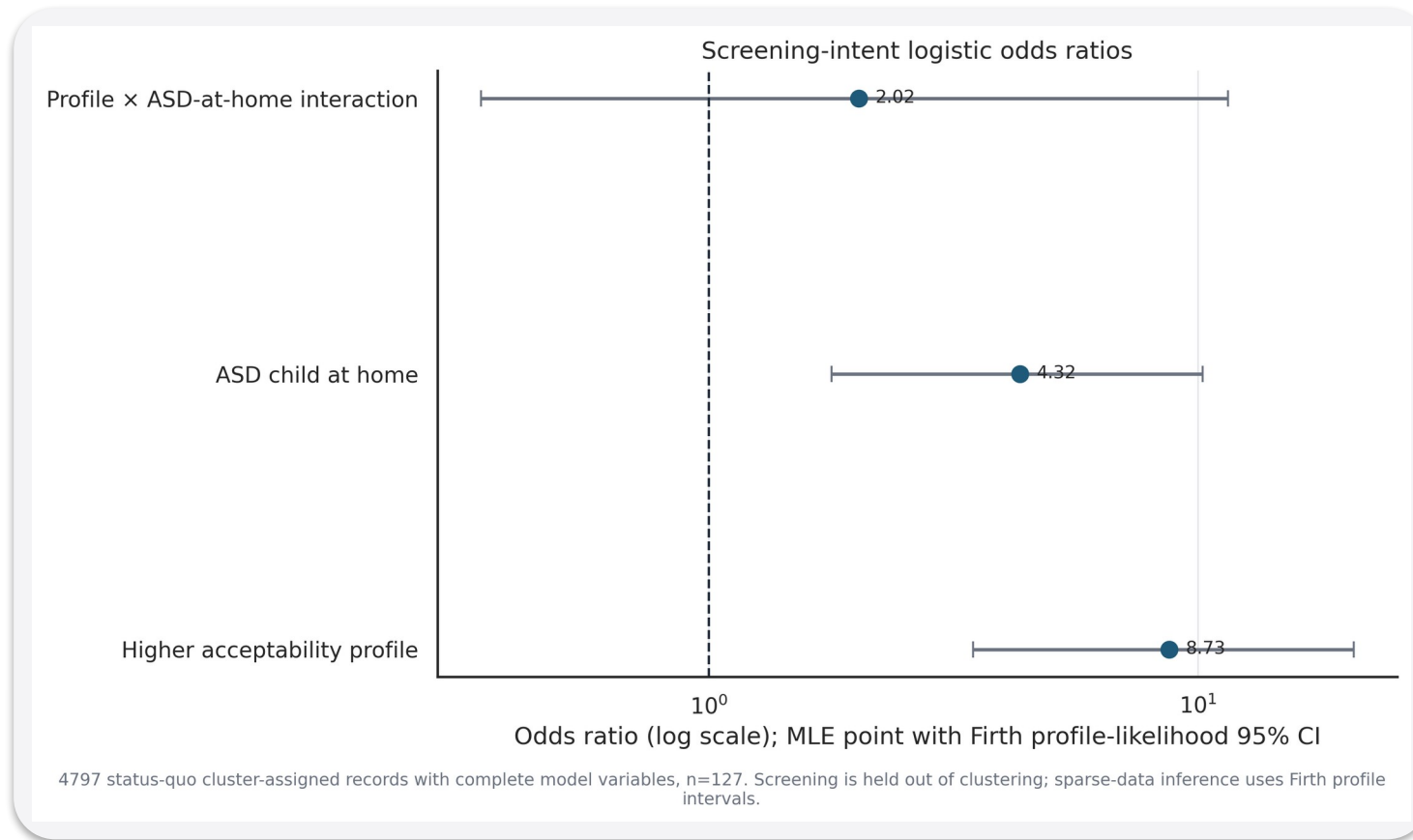
- 1 Left panel: the share who said definitely yes. The vertical bar shows how sure we are about that share.
- 2 The two bars do not overlap, which is the simple test for whether a difference is real or just noise. Here it is real.
- 3 Right panel: the full mix of answers. Notice the dark blue block is much bigger for the warmer group.

Source: figure_3_screening_outcome.png, table_5_screening_outcome.csv

Careful: this is a link, not a cause. We asked about attitudes and screening in the same sitting, so we cannot say which one drives the other.

It is not simply families who already live with autism

Having an autistic child at home does make a caregiver more likely to screen. It does not explain the group difference away.

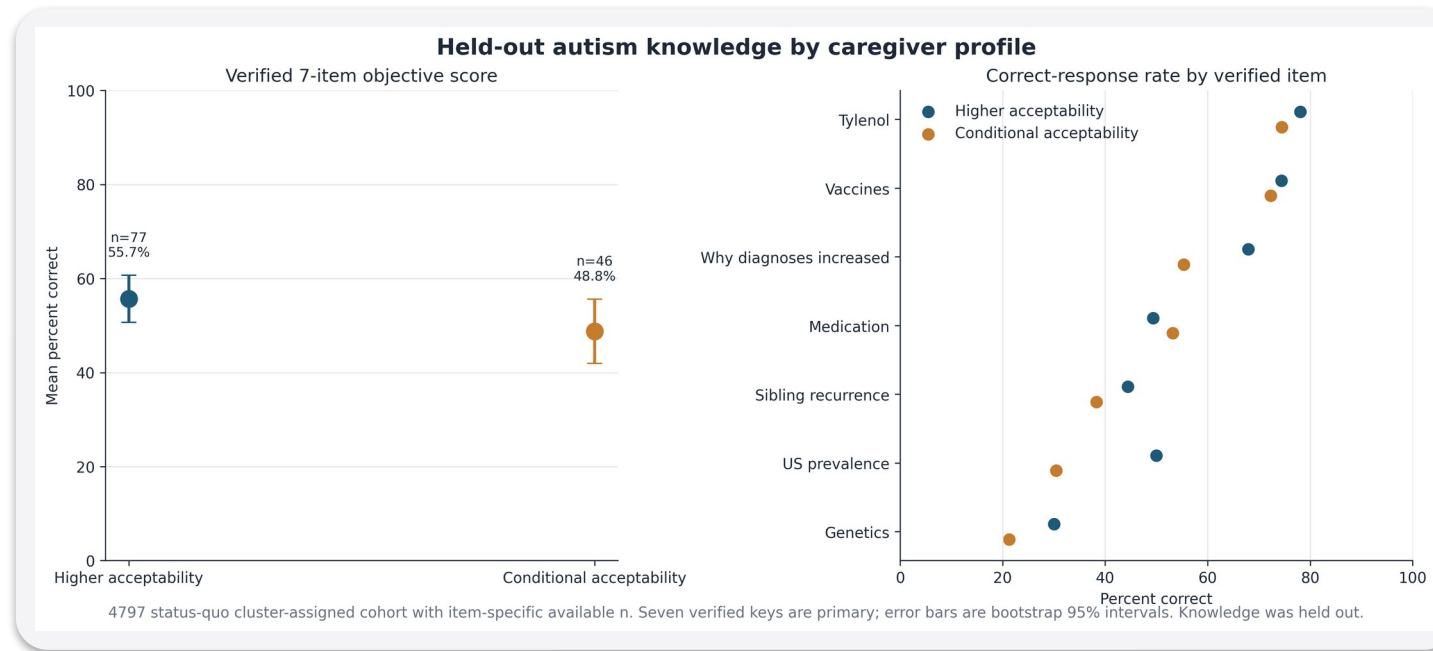


HOW TO READ THIS

- 1 Each dot is an odds ratio. Anything to the right of the dashed line at 1 means more likely to say definitely yes.
- 2 Bottom row: the attitude group is worth about 8.7 times the odds, even after we account for autism in the home.
- 3 Middle row: autism at home matters on its own, about 4.3 times the odds.
- 4 Top row asks whether the two effects feed off each other. The line crosses 1, so we cannot tell. They look like they just add up.

Some things you would expect to matter, do not

Autism knowledge, personal values and demographics all look about the same in both groups. That is a finding worth stating carefully.



HOW TO READ THIS

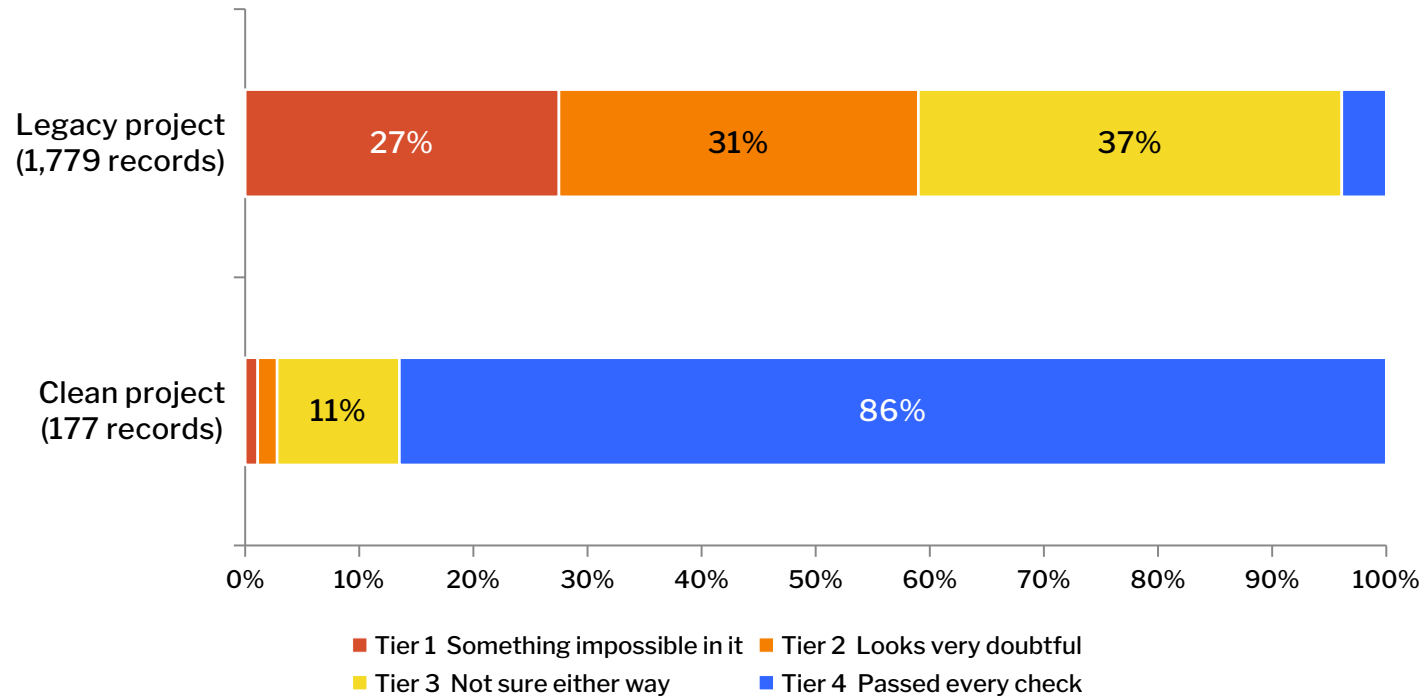
- 1** Left panel: average score on a 7 question autism quiz. The vertical bars overlap, so we cannot call this a real difference.
- 2** Right panel: the same comparison, question by question. Both groups are strong on vaccines and Tylenol, both weak on genetics.
- 3** The same picture holds for values and for demographics. Nothing survives once we account for how many things we tested.

Source: figure_4_autism_knowledge.png, figure_5_values_characterization.png, figure_6_demographic_characterization.png, table_29_power_precision.csv

We did not find a difference. That is not the same as there being none. At this sample size, gaps under about 25 points would be invisible to us.

Now the second question: how much of this data can we trust?

One project was collected under supervision. The other was flooded with low quality submissions. We score every record against the same fixed set of rules.



HOW TO READ THIS

- 1 Blue on the right is Tier 4, meaning the record passed every check. Red on the left is Tier 1, meaning something in it is impossible.
- 2 In the clean project almost 9 in 10 records pass. In the legacy project fewer than 1 in 20 do.
- 3 That is why we treat the clean project as our main dataset and the legacy one as a rough second opinion.

Source: table_16_trust_tier_counts.csv

A low tier means we do not trust the record. It does not mean a bot filled it in. We have no way to tell who or what typed the answers.

A simple check that catches impossible answers

If the survey form could not have produced an answer, the record cannot be right. There is no threshold to argue about and no judgement call to make.

A REAL EXAMPLE FROM RECORD 4581_1069

How many children do you have?

2

Which age bands are your children in?

Band 3

Band 4

Band 5

3 bands ticked

Three different age bands cannot hold two children.
The form does not allow it, so at least one answer here is not a real report.

WHY THIS ONE IS DIFFERENT

- 1 Most fraud checks say a record looks unusual. This one says the record is impossible, which is much harder to argue with.
- 2 It never flagged any of our 131 verified human caregivers, so tightening it costs no good data.

44

records flagged by this check

44

of them in the legacy project

0

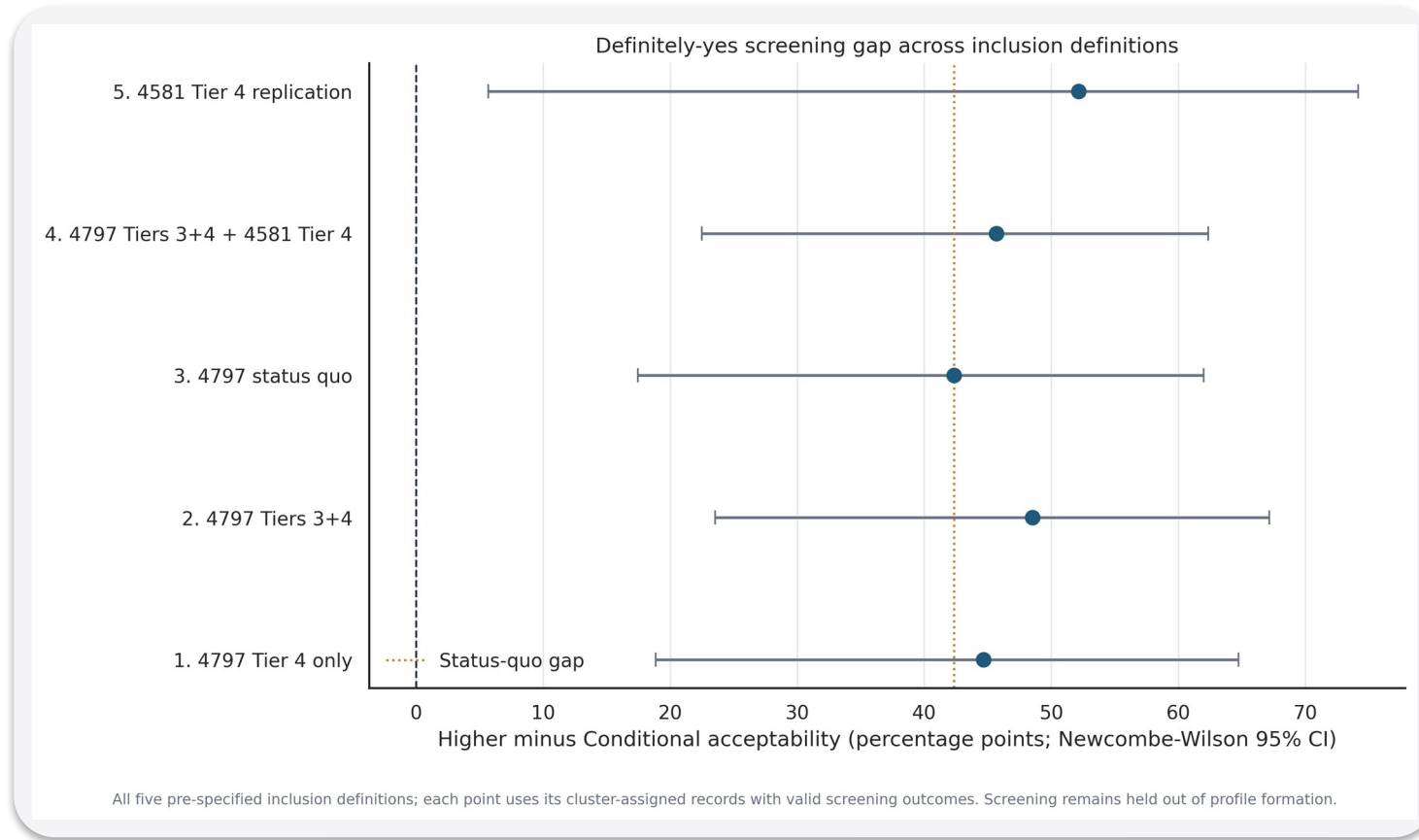
of them in the clean project

0

false alarms in 131 verified humans

Does the messy data change the main finding?

We re ran the whole analysis five times, using five different levels of strictness about which records to keep. The answer barely moved.



HOW TO READ THIS

- 1 Each row is one way of deciding which records to keep. The bottom row is the strictest, the top row uses only the legacy project.
- 2 The dot is how big the gap between the two caregiver groups is. The line shows how sure we are about it.
- 3 Every line sits well to the right of zero, so the finding holds whichever rule we pick. The gap ranges from about 42 to 52 points.
- 4 The top row looks strongest but its line is very wide, because it rests on only 47 people. Treat it as consistent rather than as proof.

What we can say, and what we will not say

Everything on the left is supported by the figures we just walked through. Everything on the right is a sentence I would ask us not to write.

We can say this

- Y The two caregiver groups are about 42 points apart on whether they would definitely screen their baby.
- Y That gap survives once we account for autism in the home, and it holds under all five strictness settings we tried.
- Y 44 records in the legacy project contain answers the survey form could not have produced.
- Y Almost 9 in 10 clean project records pass every trust check. Fewer than 1 in 20 legacy records do.

We will not say this

- N That attitudes cause screening decisions. We measured both in the same sitting.
- N That any share of records came from bots. We have no way to tell who typed them.
- N That the two groups are the same on knowledge or demographics. We just could not detect a gap this small.
- N That the 44 flagged records tell us how much of the legacy project is bad. They only tell us about those 44.

Next steps: keep the clean project as our main dataset, report trust tiers instead of labels, and use the wording on the left when we write this up.